

Video Analytics for Indoor Crowd Estimation

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Abstract—This paper demonstrates the usage of security camera footages with deep convolutional neural networks to provide cabin-level crowd density estimates in the video frames. Some applications for this include cabin-level crowd density estimates of incoming trains. With this information, train passengers may choose to board the trains at less crowded cabins, potentially decreasing the dwell time of trains at stations and experiencing a more pleasant commute overall. In a way, the crowd level estimation information will also help to maximize the train and platform capacity. Leveraging on the security camera footages would also serve as a cost-effective solution to the train operator as compared to installing new sensing equipment in the trains. Due to privacy and security concerns of publishing train cabin video frames, this paper will present the experiment results on an indoor pedestrian dataset.

I. INTRODUCTION

Overcrowding is a problem that train passengers face daily during the peak hours. A contributing factor to this problem is that the crowd distribution among train cabins are usually uneven as passengers tend to cluster at areas near the platform entrances [1]. Fig 1 shows a heat map of a train platform where the warmer colours indicate a higher crowd density. The diagram was a finding related to spatial clustering in trains and platforms, included in a report that was submitted to the Transport for New South Wales government agency. It can be seen that commuters tend to crowd in the middle rather than the end of train cars. This is very similar to the train commute experience in Singapore, and possibly around the world.

In the Internet discussion website Reddit, opinions on the best strategy to board trains were solicited by a user in the *Singapore* sub-channel, to which many mixed opinions have been received [2]. The mixed opinions are an indicator that commuters have no concrete source of information regarding the crowd levels in oncoming trains, and are choosing the queues based on intuition. A local online media company also stated that passengers were always unevenly distributed and suggests that information on crowd levels in train cabins may improve the commute experience of train passengers [3].

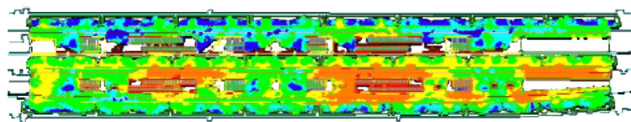


Fig. 1. Heat Map of Passengers Clustering at Train Platforms [1]

As the entrances to the train platforms are commonly located at the same area across stations, it is only natural that passengers, especially those that are rushing to board departing trains, cluster at those areas. It is also observed that platform staff are regularly being deployed to usher passengers to other parts of the platform with the intention to even the crowd distribution. However, every staff deployed is an additional resource expended for the train operator.

The motivation of this project is to facilitate a more pleasant commute by providing an estimation and visualization of the crowd levels in oncoming trains so that pre-emptive measures can be made. Such a measure may include train passengers queuing at doors where the cabin of the approaching train has a low crowd level. Additionally, the dwell time of trains may also be decreased as a result of intelligent boarding [4].

Since security cameras are already installed in every cabin of every train in Singapore [5], leveraging on the security footages would be the most cost-effective method to the train operator as compared to installing new equipment like weight sensors, infrared sensors, or thermal cameras that other crowd estimating methods may utilize.

II. RELATED WORK

A. Channel State Information in WiFi Communication Links

A device-free crowd counting method using WiFi was demonstrated by W Xi et al. in 2014 with off-the-shelf 802.11n devices. The method uses the channel state information (CSI) in WiFi communication links to predict the crowd count, and results suggested that there was a monotonic relationship between the CSI variations and number of moving people [6]. The method also has the added benefit of being immune to different lighting conditions and not requiring humans to possess any extra tracking devices. However, installation of new WiFi equipment was needed for the method to function. Furthermore, the study had only demonstrated an experiment conducted in a single monitoring area. The aim of this paper is to estimate the crowd levels in multiple train cabins individually, and it is not known how having multiple monitoring areas positioned adjacently to each other will affect the CSI readings.

B. Infrared Sensors

Lavanya et al. proposed the installation of four infrared (IR) sensors to estimate the crowd density in buses as part of a smart information system for public transport [7]. Specifically,

two IR sensors at the entrance and two IR sensors at the exit are to be installed on every bus to provide bidirectional tracking capabilities. Depending on which sensor was triggered first at a particular door, the system increments or decrements the crowd count by one. The study did not elaborate on the results of the crowd density estimation system. However, it is probable that the system cannot handle multiple passengers crossing the doors at the same time. Furthermore, this method requires the need for installation of new equipment which further raises the time and monetary cost required for implementation.

C. Background Subtraction and Texture Analysis

Suggested in [8] and used in [9], this technique estimates the crowd density by the ratio of “crowd area” to “background area” in an image. The ratio is calculated by the difference of a reference background image and the subject image. A background is determined by the pixels in a scene that are not expected to change [8]. Although this technique is conceptually simple, it may not be the most practical method to use as it relies heavily on a background image for reference. As such, challenges are introduced when the lighting conditions changes, or when external objects like new pieces of furniture are added to the background, rendering the existing reference background image to be ineffective. Such external changes to the background scenes are very likely to happen in train cabins due to the periodical changes of advertising collaterals, as well as change in illumination as trains operate in the day (natural daylight) and night (artificial light).

Another study utilized both background subtraction and texture analysis techniques to estimate crowds as described in [10]. In that method, the source image is divided into four regions of interests (ROI) to subdue the problem with perspective distortion. Then, background subtraction was used to determine if an ROI is crowded or not according to Polus’s crowd classification table [11]. Background subtraction technique is used again for low-density ROIs, and texture analysis is used for high-density ROIs. Lastly, the density calculation of all ROIs is added up to determine the final result. Due to the nature of using background subtraction techniques, the same disadvantages applies

D. Artificial Neural Networks

While there are no doubts that regular feedforward artificial neural networks (ANNs) can be easily trained to classify images [12], its flaws become apparent when tested against images that vary significantly from those in the training dataset [13]. For example, a model that is trained to classify numerical digits whose position are always in the centre of the image may not even recognize the same digits if appeared off-centre or resized. While a brute-force method such as scanning the entire image using a sliding window may work, it is extremely inefficient even before considering the fact that the right sized window has to be found and used. Another possible solution might be to increase the size of the dataset by applying data augmentation techniques to the training dataset.

Data augmentation techniques include perspective distortion, scaling, and rotation, just to name a few. Although data augmentation may improve the performance of the model [14], it is not necessarily the most efficient method as the number of layers in the network and computing power also has to be increased in parallel.

In the recent years, usage of convolutional neural networks (CNN) has been demonstrated to perform significantly well in image classification tasks as seen in [15]–[17]. CNNs are also more efficient whereby instead of feeding an entire image into the network like how ANNs do, CNNs uses kernels that slides over the entire image looking for specific patterns that are learnt over time via back-propagation during a supervised learning process. The prediction scores are then aggregated at the final layer, typically known as a fully-connected layer. Fully-connected layers have connections to every single neuron in the previous layer and possess an equal number of neurons as the number of classes in the classification problem. Fully-connected layers aggregate the output of every neuron in the previous layer and gives a prediction for each class.

III. IMPLEMENTATION

A. System Framework

Since the objective of this work was to estimate the crowd density level in train cabins, it was not necessary to obtain the exact count of people in a given image. Rather, a relative estimation of the crowd density is sufficient. As such, the recommendation of this paper was to train a CNN to classify four classes of crowd density pertaining to the dataset. Fig. 2 shows the overview of the system. Due to confidentiality issues, a substitute dataset was used in this work.

Using machine learning frameworks TensorFlow and TFLearn, an 8-layer CNN was constructed with efficiency and simplicity in mind to curb computing power limitations. Pooling layers reduce the spatial size of an image representation, resulting in lesser parameters and computation throughout the network. As such, a MaxPool layer was introduced as the first layer in the CNN and it was found to have significantly reduced the training duration without compromising on accuracy. In the network, all convolution layers use a stride value

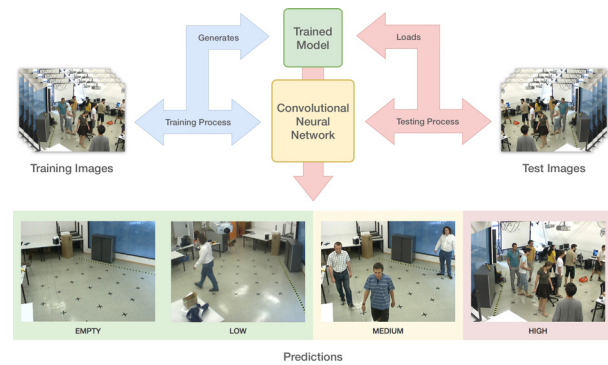


Fig. 2. Overall System Architecture

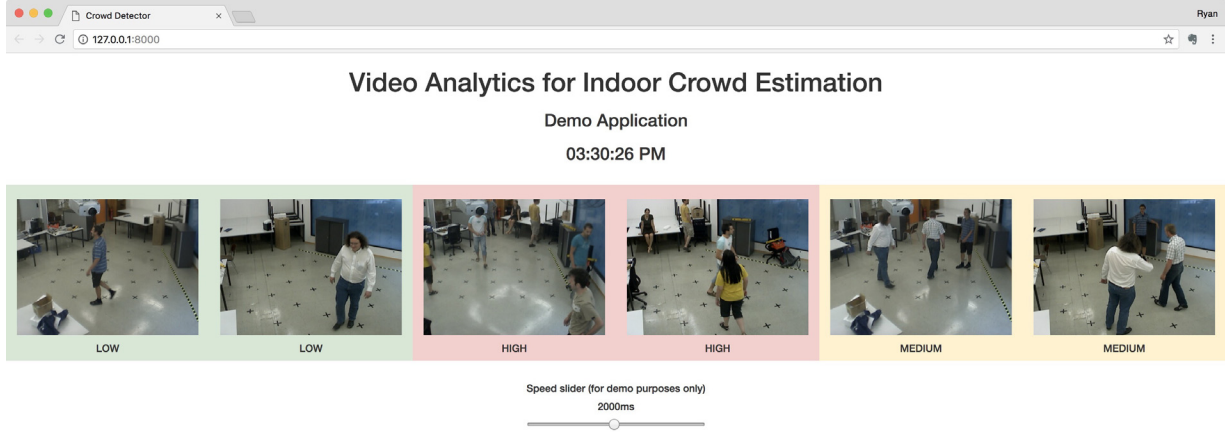


Fig. 3. Demo Web Application

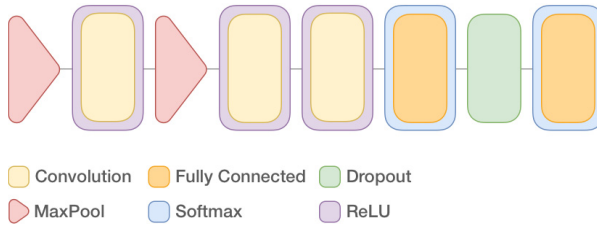


Fig. 4. CNN Architecture

of 1 as the input images are considerably small. The first two convolution layers have filters of size 7×7 , while the remaining of size 5×5 . All convolution layers uses ReLU as the activation function. The first and last convolution layers have 16 filters, while the second convolution layer has 32. All MaxPool layers have a filter size of 2×2 , with stride values of 2. A dropout layer with 50% dropout rated was inserted before the last layer in an effort to reduce overfitting, and the network ends off with a fully-connected layer that aggregates the probability of a prediction. Figure 4 shows an overview of the network architecture.

B. Visual Demonstration

To provide a visual demonstration of the crowd classification model, a web application was created with the intention to simulate an information dashboard of live crowd levels in oncoming trains with six cabins as seen in Figure 3.

Upon launching the web application, the model is loaded to the CNN. Then, six random images are selected from the test dataset. As with the testing phase, the six images are sequentially fed into the CNN for prediction. Once the prediction is done, the six images are displayed adjacently to one another in green, yellow, and red coloured boxes, representing empty and low, medium, and high crowd levels respectively. To simulate a rolling video, this process repeated on a loop every 2 seconds by default.

IV. EVALUATION

A. Dataset

The *Indoor Multi-Camera Pedestrian Datasets* by the Institute of Computer Graphics and Vision [18] was chosen as the substitute dataset for this paper. The actual train security camera dataset was not shown because of confidentiality. Although other datasets of crowds exist on the Internet, this dataset was chosen due to the resemblance of a train security camera's viewpoint, where the camera is placed at a high position in an indoor environment where the background is filled with distractions (advertisements and handrails).

As depicted in Fig. 6, the definition of each class is as follows — Empty: 0 persons, Low: 1 person, Medium: 3-5 persons, High: 5+ persons [18]. The dataset also consists of images from three camera angles which was beneficial in the training and evaluation of the model.



Fig. 6. Indoor Multi-Camera Pedestrian Dataset

Table I shows the breakdown of the dataset according to training, validation, and testing data.

TABLE I
DATASET BREAKDOWN

	Training	Validation	Testing
Empty	6075	675	750
Low	11877	1317	1462
Medium	9055	1003	1116
High	5282	586	651

The original state of the dataset, however, was not ideal for immediate utilization for two reasons. Firstly, due to the

		PREDICTION				PERFORMANCE MEASURES		
GROUND TRUTH		Empty	Low	Medium	High	Accuracy	Precision	Recall
	Empty	100%	0	0	0	99.95%	99.73%	100.00%
	Low	0.14%	94.94%	4.92%	0	97.51%	98.23%	94.94%
	Medium	0	2.24%	97.76%	0	96.76%	91.30%	97.76%
	High	0	0	4.92%	95.08%	99.20%	100.00%	95.08%
						OVERALL		
						Accuracy	Precision	Recall
						98.35%	97.32%	96.95%

Fig. 5. Confusion Matrix and Performance Measures for the Combined Camera Angles Model

original file structure of the dataset, conducting evaluations was challenging because, at the top of the hierarchy, the images are sorted according to density. Secondly, in every density, there exists images that would be classified as another class. For example, in the high crowd density dataset, there exists images with less than 5 people, which would deem those images as not belonging to a high-level dataset according to the class definitions. As such, the dataset had to be re-sorted manually.

It is not possible to feed in raw image data into a CNN (or any form of artificial neural networks) for prediction because raw data cannot be logically processed and computed. As such, the images were converted into an image matrix representation saved as a NumPy array. During the conversion, all images were downsampled to a height of 64 pixels, retaining the original aspect ratio. Due to the computing limitations of the training machine, this image size was chosen as it was big enough to contain valuable information, yet small enough to maintain a decently low training time. Ideally, the network should be built to handle larger images that would contain more valuable information for classification. However, this configuration was sufficient to demonstrate the goals of this paper. Additionally, the images are also converted to greyscale in an effort to prevent any colour-bias forming in the model during the training process.

B. Experimental Setup

Given the three camera angles in the dataset, four sets of evaluations were conducted — the first three for each camera angle individually, and the fourth for all cameras angles combined. The goal of this experiment was to see if the model was effective in predicting single or multiple scene datasets. The hypothetical recall rate threshold for a model to be deemed "capable" is set at 90%, and it is ideal that the dataset with combined camera angles surpasses this threshold as it would mean that the model is capable in estimating the crowd level, regardless of the background environment. For each experiment, a new model was trained and evaluated with its own training, validation, and testing dataset.

C. Results and Discussion

As seen in Table II, the recall rate of all models had surpassed the hypothetical recall rate threshold of 90%. It

TABLE II
EVALUATION RESULTS

	Accuracy	Precision	Recall
Model 1	99.89%	99.70%	99.85%
Model 2	99.73%	99.55%	99.49%
Model 3	99.17%	98.57%	98.28%
Model 4	98.35%	97.32%	96.95%

is also worth noting that the confusions of the model were adjacent to the true positive line whereby an image of class "Low" was either falsely predicted as "Empty" or "Medium", but not "High". Model 4, the combined camera angles model, achieved a recall rate of 96.95%. As discussed previously, the ideal model would be the one that was trained and tested with the combined camera angles as it shows that it is capable of estimating the crowd level regardless of the background environment. Since model 4 has fulfilled the conditions, it was chosen as the final model to be used for the project. Fig. 5 shows the confusion matrix for model 4.

During the training process, fastest prediction rate achieved was 843.91 predictions per second or approximately 212 milliseconds per image. It is crucial that the prediction speed is kept as fast as possible to provide the most current crowd level estimations for multiple train cabins simultaneously. As the machine used for the training and testing of the model was not of ideal specifications, it is inferable that given a machine with better performance, the prediction speed of the model could be increased significantly.

V. CONCLUSION

This paper demonstrates a video analytics solution for crowd estimation in train cabins. The solution was to process images extracted from security camera footages with a deep convolutional neural network to estimate the crowd density. The system comprises two phases: the training phase, and the testing (or production) phase. In the training phase, a total of 35,870 labelled images were fed into the network to conduct supervised learning. Thereafter, the model was tested against 3979 unseen images and achieved an overall recall rate of 96.95%.

A. Future Work

1) *Better Hyper-Parameter Tuning*: The hyper-parameter values of the current model were tuned on a trial-and-error basis within a limited development time frame. This was because the training process was performed on a machine with considerably low specifications which further prolonged each training process. To overcome this challenge, methods like downscaling the images in the dataset and introducing a pooling layer as the first layer of the network were put in place. Even then, the training of each model took several hours which was not helpful considering the limited time frame of the project. To further enhance the robustness and scalability of the model, more research on how to build the model for better accuracy and dynamic input data could be put in place.

2) *Web Service Integration to Aid Front-End Devices*: This crowd classification algorithm may be further developed and integrated as a web service where multiple front-end applications may retrieve information from. Such front-end devices include mobile applications for commuters and other Internet of Things devices. A more ambitious approach would be to install lights at the platforms that signal green, amber, and red to indicate the crowd densities of the oncoming train.

3) *Adoption in Bus Services*: Buses in Singapore currently uses infrared sensors to count passengers in the upper deck of buses [19]. Since security cameras are also present in buses in Singapore, additional research could be done to see if the crowd classification model used in this project could outperform the infrared sensors used in the buses, or if not, how to tweak the model to work in buses.

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